

**END SEMESTER ASSESSMENT (ESA) - Jan - May 2024****UE18CS353 - Object Oriented Analysis and Design with Software Engineering****Total Marks : 100.0**

1.a. Using a skeletal system block diagram, compare and contrast Iterative and Incremental Development Models. (6.0 Marks)

1.b. What is the problem with the following requirement statement? That is, what quality attributes the statement does not possess? How can it be improved?

**“SR-4: All pages of the application will load in acceptable amount of time.”**  
(6.0 Marks)

1.c. Describe what is a risk and the steps (with 1 line description) involved in managing risks. (8.0 Marks)

2.a. Describe briefly (1-2 lines) any six approaches that can be taken to decompose a large system into subsystems or components.

(6.0 Marks)

2.b. Draw the Class Diagram to model a system for management of courses and enrolments. Identify classes, attributes, relationships and cardinality.

1. A Department manages multiple Sessions.
2. A Session has a Start date, an End date and Name (like Jan-May 2023).
3. A session can have multiple semesters running (like 6<sup>th</sup>, 4<sup>th</sup>, 2<sup>nd</sup> etc).
4. A student belongs to a semester and to a section within it. A section has minimum of 60 students.
5. A semester has a set of courses offered in it.
6. A student has to enrol for all the required number of courses (let us assume 5 for the listed semesters). For some courses like electives, a minimum of 20 students have to enrol for it to be offered.
7. An enrolled student's attendance for the course is recorded.
8. Similarly, an enrolled student's ISA, ESA marks and Grade for the course are recorded.
9. A teacher will be assigned to a section for a course.
10. A teacher can also be assigned as the class teacher for a particular section

Note: Assume attributes and other details as appropriate where not provided.

(8.0 Marks)

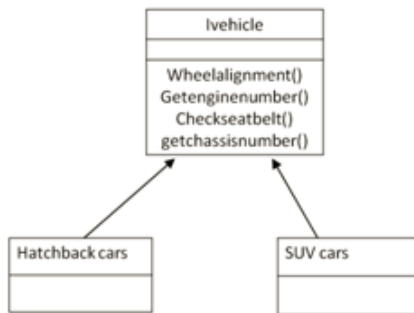
2.c. For the given scenarios, choose appropriate diagram to model the scenario. Justify your choice and draw the chosen diagram.

- i) Every student is assigned a Faculty Advisor. The Faculty Advisor is assigned a minimum of 25 students with no maximum limit. Deans and Chairpersons are not assigned any students as advisees. Typically, about half the strength of a section is assigned to a Faculty advisor, i.e., a section typically has two Faculty Advisors.
- ii) An Order consists of multiple Items. The Order total is computed by adding the product of Item Quantity and Item price for each Item.

(6.0 Marks)

3.a. Class design given below has functionalities of vehicles. Now, the client wants to add a two wheeler class to the design. If the new two wheeler class implements interface Ivehicle, which design principle is violated? Modify the design in the given class diagram to accommodate the new class.

(8.0 Marks)



3.b. Differentiate between Coding standards and guidelines for an organization with at least 2 examples for each.

(6.0 Marks)

3.c. What is secure programming? List the secure programming approaches or practices to overcome software vulnerabilities.

(6.0 Marks)

4.a. Choose most appropriate types of testing applicable in the following applications from the list of testing types given below:

**Functional, System, Load, Stress, Performance, Compatibility, Security, Reliability, Availability, Regression, Usability, Acceptance**

1. Online shopping website (like Amazon), where they offer special sale days (like Prime Days).
2. A file management system (like Google Drive), where different types of files can be uploaded and viewed from multiple devices. It also needs to provide collaborative editing feature.
3. Anti-Virus software that needs to be kept up-to-date (in terms of software and data related to newer vulnerabilities) frequently. (6.0 Marks)

4.b. What are the 3 perspectives for looking at test related measures? Give one example for each. (6.0 Marks)

4.c. What is SEI CMM? Name two reasons why you would want to use it? Enumerate its five levels. (8.0 Marks)

5.a. List any six objectives of IT Systems Management (ITSM). (6.0 Marks)

5.b. Name any three challenges of Geographical distance with respect to Software Development? How can they be overcome? (6.0 Marks)

5.c. Describe the term DevOps. Write briefly about the DevOps pipe line. (8.0 Marks)